

Java Intro Exercise

Exercise 1 : Error Analysis

Instructions:

1. **Analyze the Code (Program 1):**
 - The following Java program contains multiple errors and issues related to Java identifiers, naming conventions, assignment operators, increment/decrement operations, primitive and class data types (implicit and explicit casting), variables and reference variables, as well as static and final initialization.
2. **Identify Errors and Issues:**
 - Without running the code, carefully examine each line and think about potential syntax and logical errors.
3. **Refer to Your Lecture Notes:**
 - Cross-check with your notes or textbook to understand what might be incorrect.
4. **Write Explanations in Comments:**
 - For each identified issue, write a brief explanation in the code comments about why it is incorrect and how to fix it.
5. **Fix the Code and Run It:**
 - Correct the errors, run the modified version, and observe the output.
6. **Reflect on Learning:**
 - Summarize the key takeaways from this exercise.

```
class Analyze1{
    static final int MAXVALUE = 100;
    final static double pi = 3.14159;

    public static void main(String[] args) {
        int 1number = 10; // variable names cant start with a
digit.
        double Double = 20.5; //cannot use java keywords as
variable names

        int a = 5;
        int b = a++;
        int c = --b;

        int smallNumber = 100;
        double largeNumber = smallNumber;
        int convertedValue = (int) largeNumber;

        String name = "Sarah";
        String anotherName = new String("Sarah");

        final int FinalVariable;
        FinalVariable = 50;

        System.out.println("Value: " + FinalVariable);
    }
}
```

Program 1

Exercise 2 : Using Scanner and JOptionPane in Java

Case Study: Calculating BMI (Body Mass Index)

Objective:

Students will learn how to:

1. Use the `Scanner` class to get input from the keyboard.
 2. Use `JOptionPane` to get user input via a dialog box.
 3. Convert a string input into a numeric type using `Double.parseDouble()`.
-

Scenario:

You are developing a simple program that calculates the Body Mass Index (BMI). The BMI is calculated using the formula:

$$BMI = \frac{\text{weight in kg}}{(\text{height in meters})^2}$$

The user should have two options to enter their weight and height:

1. **Using the keyboard (Scanner).**
 2. **Using a graphical input box (JOptionPane).**
-

Task 1: Using Scanner for Input

Write a Java program that:

- Asks the user to enter their **weight (kg)** and **height (m)** using `Scanner`.
- Computes the BMI and displays it.
- Analyze and improve the code given here :

```
import java.util.Scanner;

public class BMIScanner {
    Public Static main(string[] args) {
        Scanner input = new Scanner(System.in);

        // Get user input
        System.out.print("Enter your weight (kg): ");
        double weight = input.nextDouble();

        System.out.print("Enter your height (m): ");
```

```

        double height = input.nextDouble();

        // Calculate BMI
        double bmi = weight / (height * height);

        // Display result
        System.out.println("Your BMI is: " + bmi);

        input.close();
    }
}

```

Task 2: Using JOptionPane and Parsing Input

Modify the program to:

- Use `JOptionPane.showInputDialog()` to get the weight and height as **String**.
- Convert the string input to double using `Double.parseDouble()`.
- Compute and display the BMI using `JOptionPane.showMessageDialog()`.

```

import javax.swing.JOptionPane;

public class BMIJOption {
    public static void main(String[] args) {
        // Get input as String
        String weightStr = JOptionPane.showInputDialog("Enter your weight (kg):");
        String heightStr = JOptionPane.showInputDialog("Enter your height (m):");

        // Convert input to double
        double weight = Double.parseDouble(weightStr);
        double height = Double.parseDouble(heightStr);

        // Calculate BMI
        double bmi = weight / (height * height);

        // Display result
        JOptionPane.showMessageDialog(null, "Your BMI is: " + bmi);
    }
}

```

Task 3 : Student Exercises

1. **Run both versions** of the BMI calculator and compare their behavior.
2. **Modify the program** to display the BMI **category** (Underweight, Normal, Overweight, Obese).

```

import javax.swing.JOptionPane;

```

```
public class BMIJOption {  
  
    public static void main(String[] args) {  
  
        String weightStr = JOptionPane.showInputDialog("Enter your weight (kg):");  
  
        String heightStr = JOptionPane.showInputDialog("Enter your height (m):");  
  
  
        double weight = Double.parseDouble(weightStr);  
  
        double height = Double.parseDouble(heightStr);  
  
  
        double bmi = weight / (height * height);  
  
        String category = BMICategoryHelper.getBMICategory(bmi);  
  
  
        JOptionPane.showMessageDialog(null, "Your BMI is: " + bmi + "\nCategory: " + category);  
  
    }  
  
}
```

This tutorial helps students understand **console-based** and **GUI-based** input methods while reinforcing **parsing and type conversion** in Java.